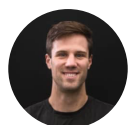


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August 4, 2025

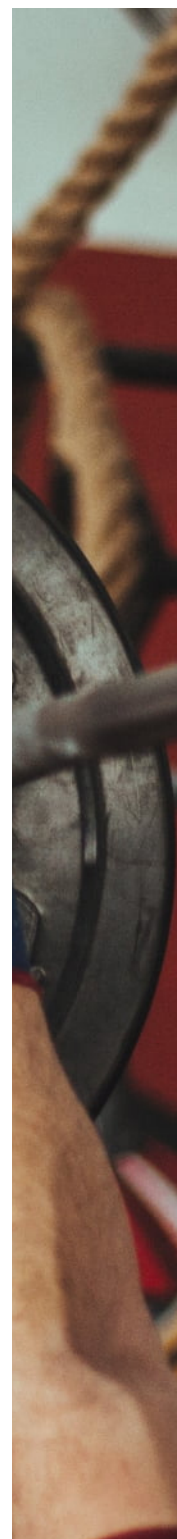
# The science that makes Velocity Based Training work

Explore the cutting edge of strength and conditioning with this guide to Velocity Based Training. Learn the science behind VBT and its impact on performance.



Jacob Tober  
Co-founder

In recent years, the field of strength and conditioning has seen a surge in popularity, with new training methods and technologies being developed every day. One such innovation is Velocity Based Training (VBT), which has taken the fitness world by storm. But what exactly is VBT, and how does it work? In this article, we will explore the science



behind this cutting-edge training method and its impact on athletic performance.

## The science behind Velocity Based Training

**Velocity Based Training** is a novel approach to strength and conditioning that takes into account an individual's movement speed in order to track progress and optimise performance. It is a data-driven approach that uses some form of sensors (camera, string, accelerometer) and software to measure bar velocity, force, power, and other key performance metrics. This information is then used by coaches and lifters to prescribe training intensity, volume, and more in real-time, providing athletes with immediate feedback on their performance.

The ultimate goal of doing this is that training can be more finely tuned to respond to the multiple factors that influence performance on a day-to-day basis. This includes, fatigue, readiness, motivation, peaking and tapering, injury, and faster or slower rates of adaptation.

Optimal  
training  
isn't some  
vague  
concept,  
it's an  
achievable  
goal.  
Training  
optimally  
leads to  
fewer  
injuries,

better  
progress  
and less  
wear and  
tear on the  
body.

## How Velocity Based Training works

VBT is rooted in the principles of physics, biomechanics, and exercise science. The velocity of a movement is determined by the speed at which the weight is lifted and the time it takes to complete the lift. By tracking the velocity of an athlete's movements, VBT provides an objective measurement of their strength and power output, allowing coaches and trainers to identify areas for improvement and tailor training programs accordingly.

The velocity of a movement is influenced by several factors, including the weight lifted, the length of the muscle fibres, type of muscle contraction, and the speed of nerve impulses. VBT provides objective insights into an athlete's motor skills, muscle imbalances, and technical weaknesses, and strengths. This information can then be used to develop customised training programs that are tailored to the individual's needs, ensuring that they are training in a safe and effective manner.

VBT has been shown to have a positive impact on athletic performance in several ways. Firstly, it provides athletes with real-time feedback on their performance, allowing them to make immediate improvements. Secondly, by tracking velocity, force, and power, VBT provides an objective measurement of an athlete's strength and power output, allowing coaches to identify areas for improvement and

adjust training programs accordingly.

Weakley,  
2021. In-  
session  
velocity  
feedback  
leads to  
greater  
athletic  
transfer and  
adaptations  
from  
training.

VBT also promotes proper lifting technique and reduces the risk of injury. By tracking bar velocity, VBT can identify areas where an athlete is losing speed or slowing down, which may indicate poor form or low readiness. This information can then be used to address these issues or adjust the days training up or down to match this readiness state.

Finally, VBT has been shown to improve athletic performance by optimising training intensity and volume. By tracking an athlete's performance in real-time, VBT can provide immediate feedback on the effectiveness of the training program, allowing coaches to adjust intensity and volume as needed to ensure that the athlete is pushing themselves to their maximum potential.

Dorrell,  
2020.  
Training

with a  
velocity  
driven  
program  
leads to  
more  
strength  
gains than  
traditional  
percentage  
based  
approaches

## Uses of velocity based training

- Provides real-time feedback on performance
- Offers an objective measurement of strength and power output
- Identifies areas for improvement and technical weaknesses
- Promotes proper lifting technique and reduces the risk of injury
- Improves athletic performance by optimising training intensity and volume

## FAQs about Velocity Based Training

### What equipment is needed for Velocity Based Training?

VBT requires the use of a tracking sensor and software. In the past this required expensive barbell velocity tracking hardware such as linear positional transducers, force plates, accelerometers, 3D motion

camera systems, and other data-acquisition devices were needed, making velocity tracking prohibitive.

Today, modern smartphones have enough computer processing power and high resolution cameras to provide accurate velocity data with just the device you already have in your pocket. Metric VBT is a one-of-a-kind app that does just this, you can download it [here](#) and use it in your training for free.

## **Is Velocity Based Training only for advanced athletes?**

No, VBT can be used by athletes of all skill levels, from beginners to professionals. The beauty of VBT is that it provides an objective measurement of an individual's performance, allowing coaches and trainers to tailor training programs to their specific needs.

## **How does Velocity Based Training compare to other training methods?**

VBT offers several advantages over traditional training methods. It provides real-time feedback on performance, allows for objective measurement of strength and power output, and optimises training intensity and volume. It also reduces the risk of injury by promoting proper lifting technique.

## **Can Velocity Based Training be used in conjunction with other training methods?**

Yes, VBT can be used in conjunction with other training methods, such as weightlifting, powerlifting, and bodybuilding. It can provide valuable information on an athlete's performance and help to optimise their training program. Velocity based training works nicely alongside other programming systems too, including RPE / RIR training, percentage based training, linear models, high volume training and much more.

## Conclusion

Velocity Based Training is a cutting-edge approach to strength & conditioning, barbell training, powerlifting and general strength training that provides real-time feedback on performance and allows for objective measurement of strength and power output. With its roots in the principles of physics, biomechanics, and exercise science, VBT is a scientifically-validated method for improving athletic performance and reducing the risk of injury.

Whether you're a beginner or a professional athlete, VBT can provide valuable insights into your performance and help you reach your full potential, and with Metric VBT you can get started now!

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...reduced analysis and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.



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